

Pranav Khadse

+91 9370768248 | psjkhadse@gmail.com | Gandhinagar, India | [LinkedIn](#) | [GitHub](#) | [Behance](#)

EDUCATION

Vellore Institute of Technology, Chennai <i>B.Tech in Electronics and Computer Engineering — CGPA: 8.74/10</i>	2023 – 2027 Chennai, India
Dr. Khatri College of Science <i>Senior Secondary (XII), Maharashtra State Board</i>	2023 Chandrapur, India
Mount Carmel Convent High School <i>Secondary (X), CBSE — Percentage: 93.8%</i>	2021 Chandrapur, India

EXPERIENCE

Embedded Programming Lead <i>Namo Nirvana – Robotics Team</i>	Mar 2024 – Mar 2026 Chennai, India
- Worked on embedded systems and firmware development for ESP32, STM32, Raspberry Pi Pico, using Embedded C, C++ and exposure to MicroPython. - Designed and validated PCBs using EasyEDA, including MCU programming, debugging, and testing.	

CERTIFICATIONS

IBM DevOps Certification <i>IBM Career Education Program (IBMCE)</i>	May 2025 – Jun 2025 Virtual
Gained knowledge of DevOps practices, Agile methodologies, CI/CD pipelines, collaboration workflows, and design thinking principles.	

PROJECTS

Secure BLE Communication System with Application-Layer Encryption <i>ESP32-C3, BLE, AES-128</i>	Apr 2026 – May 2026
- Built a secure BLE communication system on dual ESP32-C3 devices with application-layer AES-128 encryption. - Implemented dynamic key synchronization, packet fragmentation/reassembly, and secure message transmission. - Designed a menu-driven OLED + keypad interface supporting PIN/key updates, receiver monitoring, and retransmission.	
AWS-Based IoT Patient Monitoring System <i>AWS, ESP32, IoT</i>	Sep 2025 – Jan 2026
- Built a serverless patient monitoring system using ESP32 and biomedical sensors streaming vitals to AWS IoT Core. - Processed data via AWS Lambda, stored in DynamoDB, triggered SNS alerts, and visualized metrics via a static S3 dashboard.	
DevOps(CI/CD) Integrated Sensor Data Analyzer <i>ESP32, Docker, Jenkins, LoRa</i>	May 2025 – Jun 2025
- Built a DevOps-integrated sensor analyzer using ESP32 & LoRa with CI/CD pipelines (Git, Jenkins, Maven). - Implemented Docker containerization, Puppet configuration management, and real-time monitoring using Graphite & Grafana.	
AURA – Aquatic Unmanned Research Assistant (ROV)	May 2024 – Jan 2025
- Designed a modular underwater ROV integrating motor control, environmental sensing, and real-time telemetry. - Developed a Flask backend with WebSocket streaming and a React-based control UI for low-latency navigation.	

TECHNICAL SKILLS

Languages: C/C++, Java, Python, Embedded C
Frameworks & Tools: Flutter, Git, Docker, Jenkins, Maven, Oracle SQL, Puppet
Domains: Software Development, Embedded Systems, IoT, DevOps (CI/CD), AWS Cloud Computing
Core Concepts: Data Structures, DBMS, Computer Networks, OOP, SDLC,

EXTRACURRICULAR

Designing Lead <i>Game Development Club, VIT Chennai</i>	2024 – Present Chennai
Organized hackathons/workshops and designed posters, banners, and technical events.	

ACHIEVEMENTS

- 3rd Prize – Gujarat Robofest 4.0:** Underwater Robot & Submarine category among 1,284 teams nationwide.
- 1st Prize – BIS Project Showdown, VIT Chennai**
- Published Patent Application:** *Modular Control Board with Dual-MCU Architecture for ROV Navigation and Data Processing*, Publication No. IN202541039559 A1. Application No. 202541039559